New Jersey
Student Learning
Assessments-Adaptive

Practice Test

Mathematics

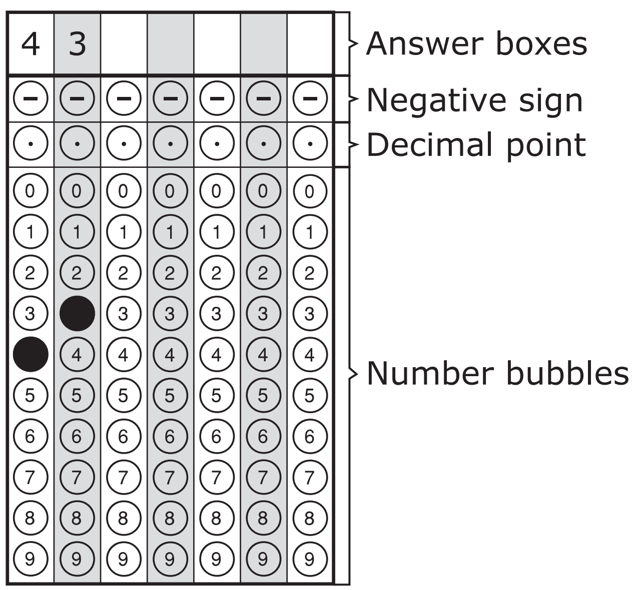
Grade 6

Student Name ____________________

Directions:

Today, you will take the Grade 6 Mathematics Practice Test. This test has two segments. In the first segment, you may not use a calculator. In the second segment, you may use a calculator. If you do not know the answer to a question, you may go on to the next question. When you finish the first segment, you may review your answers and any questions you did not answer in the first segment ONLY. When you finish the second segment, you may review your answers and any questions you did not answer in the second segment ONLY.

1. For questions with bubbled responses, fill in the circle next to your answer choice. If you change your answer, make sure you erase your old answer completely. Do not cross out or make any marks on the other choices.
2. For questions with response boxes, write your answer neatly, clearly, and in the space provided.
3. For questions with response grids, write your answer in the answer boxes at the top of the grid.
   - Write only one digit or symbol in each answer box.
   - Use only the options listed below each box.
   - Be sure to write a negative sign or decimal point in the answer box if it is part of the answer.



Segment 1

(Non-Calculator)

Calculators may NOT be used in this segment

1. Select all of the expressions that are equivalent to 3.1^4.

Ⓐ (3.1)(3.1)(3.1)(3.1)

Ⓑ (9.61)(3.1)(3.1)

Ⓒ (3.1)(3.1)(2)

Ⓓ (9.61)(9.61)

Ⓔ (3.1)(4)

2. Charlie uses the expression $2x + 2x$ to represent the perimeter of a square.

What does the variable x represent in this expression?

Ⓐ the area

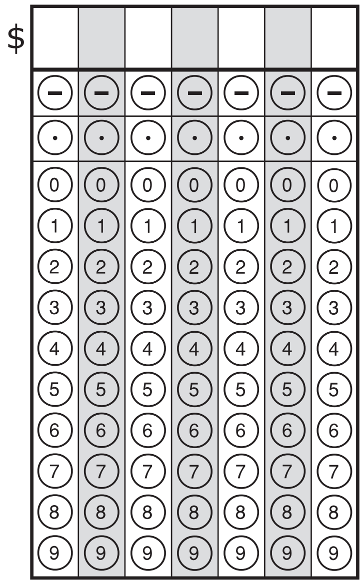
Ⓑ the perimeter

Ⓒ the side length

Ⓓ the sum of 2 side lengths

3. A school earns \$270 from selling 50 tickets for the school play. Each ticket costs the same price.

How much does the school earn for 1 ticket?



4. The total number of hours, *t*, that Trent has practiced his guitar after *d* days is modeled by the equation shown.

$$t = 3d$$

Complete the table to describe this relationship.

d	***t***
1	
3	
	15

5. Max builds a model of his house. He spends \$48.86 on materials for the model as shown:

- 3 packages of wood at \$5.99 each
- 10 miniature windows at \$1.79 each
- a roofing package for *r* dollars

Create an equation that could be used to solve for the cost, *r*, in dollars, of the roofing package.

6. On Monday, the theater club sells 14 tickets for the school play. On Tuesday, they sell 10 tickets. On those two days, they sell 32% of the total number of tickets.

What is the total number of tickets?

7. Which statement about the comparison of $|-3.2|$ and $|2|$ is accurate?

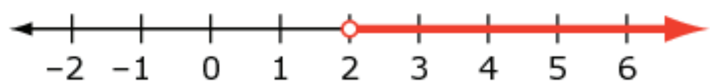
Ⓐ $|-3.2|$ is less than $|2|$ because -3.2 is a negative number.

Ⓑ $|-3.2|$ is greater than $|2|$ because -3.2 is a greater distance from 0.

Ⓒ $|-3.2|$ is less than $|2|$ because the distance between the values on the number line is greater than $|-3.2|$.

Ⓓ $|-3.2|$ is greater than $|2|$ because the distance between the values on the number line is smaller than $|-3.2|$.

8. The number line shows all of the possible values of *m*.

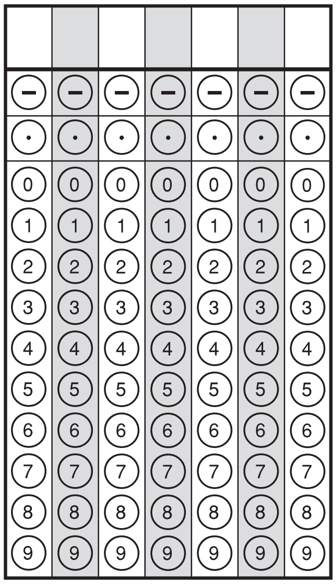


Create an inequality that represents all of the possible values of *m*.

GO ON ▶

9. Sarah types a 3-page report in 30 minutes, and each page has 400 words.

What is Sarah's typing rate, in words per minute?



10. Select all of the expressions that are equivalent to $10a + 6b$.

Ⓐ $16ab$

Ⓑ $2(5a + 3b)$

Ⓒ $10(a + 6b)$

Ⓓ $6(a + b) + 4a$

Ⓔ $5(2a) + 3(2b)$

11. Which statement is true about the points represented by (1, −3) and (2, 3)?

Ⓐ The points are located above the *x*-axis.

Ⓑ The points can be connected by a vertical line.

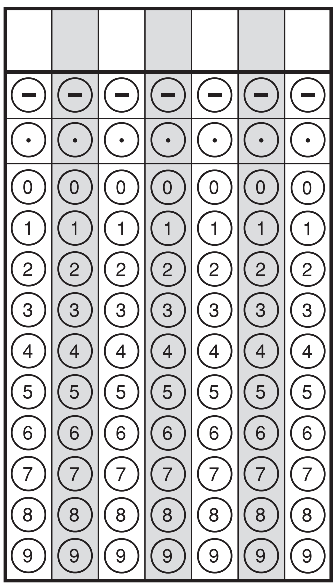
Ⓒ The points are the same distance from the *x*-axis.

Ⓓ The points are the same distance from the *y*-axis.

12. An expression is shown.

$$\frac{1}{2}b^2 - 6$$

What is the value of the expression when $b = 4$?



13. A butcher displays the cost, in dollars, of different amounts of meat in the table shown.

Cost of Meat

Amount of Meat (pounds)	Cost (dollars)
3	21
5	35
7	49

Create an equation that can be used to find the cost, *c*, for *p* pounds of meat.

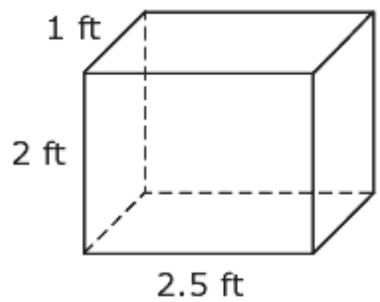
STOP

Segment 2

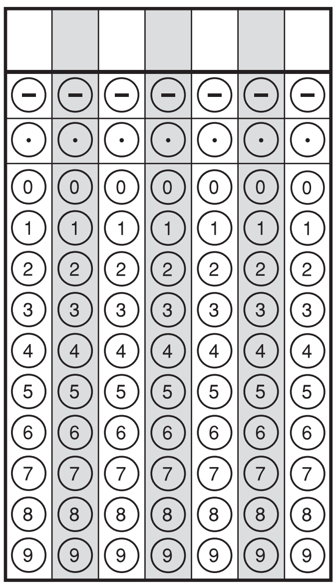
(Calculator)

Calculators MAY be used in this segment

14. A storage container is shown, with measurements in feet (ft).



What is the volume, in cubic feet, of the storage container?



15. Lee records the height of players on a basketball team, and calculates the mean absolute deviation of the set of data.

What does the mean absolute deviation of the set of data describe?

Ⓐ the average height of a player on the team

Ⓑ the height of the tallest player on the team

Ⓒ the difference in height between the shortest and tallest player

Ⓓ the average difference in height between each player and the mean height

16. Kaden works 5 days.

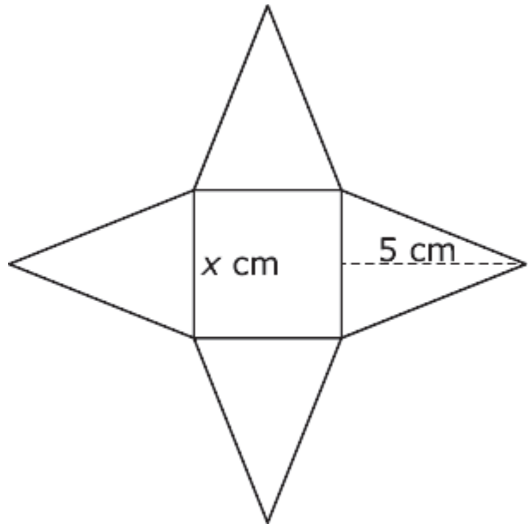
- The median number of hours he works is 3.
- The mean number of hours he works is 4.

Complete the table to show a possible number of hours that Kaden works on each of the 5 days.

Kaden Working

Day	Hours
Monday	
Tuesday	
Wednesday	
Thursday	
Friday	

17. The net of a square pyramid is shown.



Create an expression that represents the surface area, in square centimeters (cm), of the square pyramid.

STOP

Paper Practice Test Answer and Alignment Document: Math

Grade 6 Mathematics

Non-Calculator Section

Item Sequence	Answer Key	Standards Alignment
1	A, B, D	6.EE.A.1
2	C	6.EE.B.6
3	5.4, or any equivalent value	6.RP.A.2
4	Row 1: 3 Row 2: 9 Row 3: 5 or any equivalent values	6.EE.C.9
5	$3(5.99) + 10(1.79) + r = 48.86$, or any equivalent equation	6.EE.B.7
6	75, or any equivalent value	6.RP.A.3c
7	B	6.NS.C.7c
8	$m > 2$, or any equivalent inequality	6.EE.B.8
9	40, or any equivalent value	6.RP.A.2
10	B, D, E	6.EE.A.3
11	C	6.NS.C.6b
12	2, or any equivalent value	6.EE.A.2c
13	$c = 7p$, or any equivalent equation	6.EE.C.9

Calculator Section

Item Sequence	Answer Key	Standards Alignment
1	5, or any equivalent value	6.G.A.2
2	D	6.SP.A.3
3	Any set of 5 numbers with a mean of 4 and median of 3	6.SP.B.5c
4	$4\left(\frac{5x}{2}\right) + x^2$, or any equivalent expression	6.G.A.4